UML DESIGN DOCUMENT

Generative AI-Based Cloud Removal and Reconstruction for LISS-IV Satellite Imagery

System analysis and design using UML diagrams

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FIGURE 1

Use Case Diagram

Actors and the functions they perform in the cloud removal system. The administrator trains the generative model; the analyst runs scenes through the pipeline.

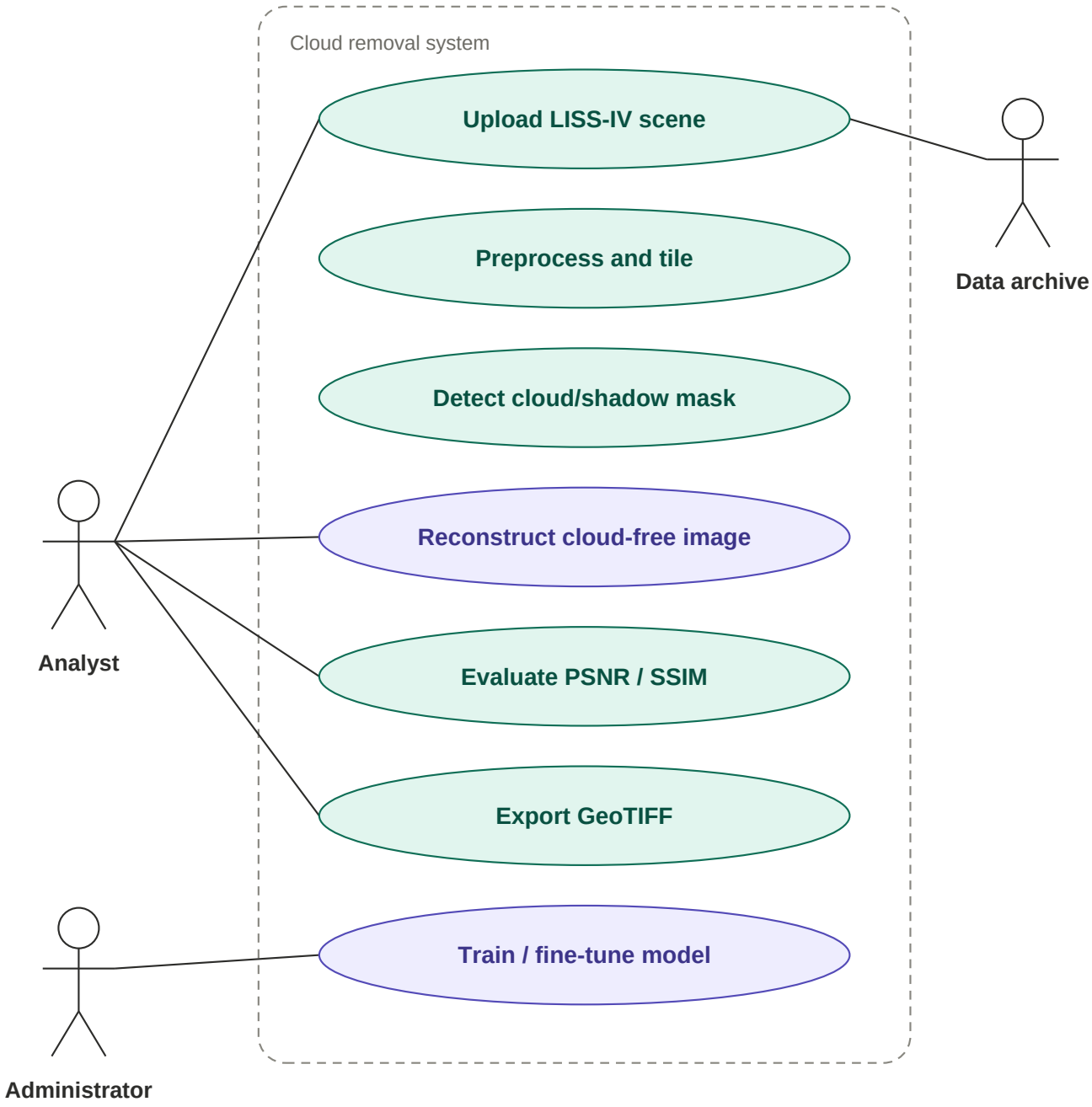
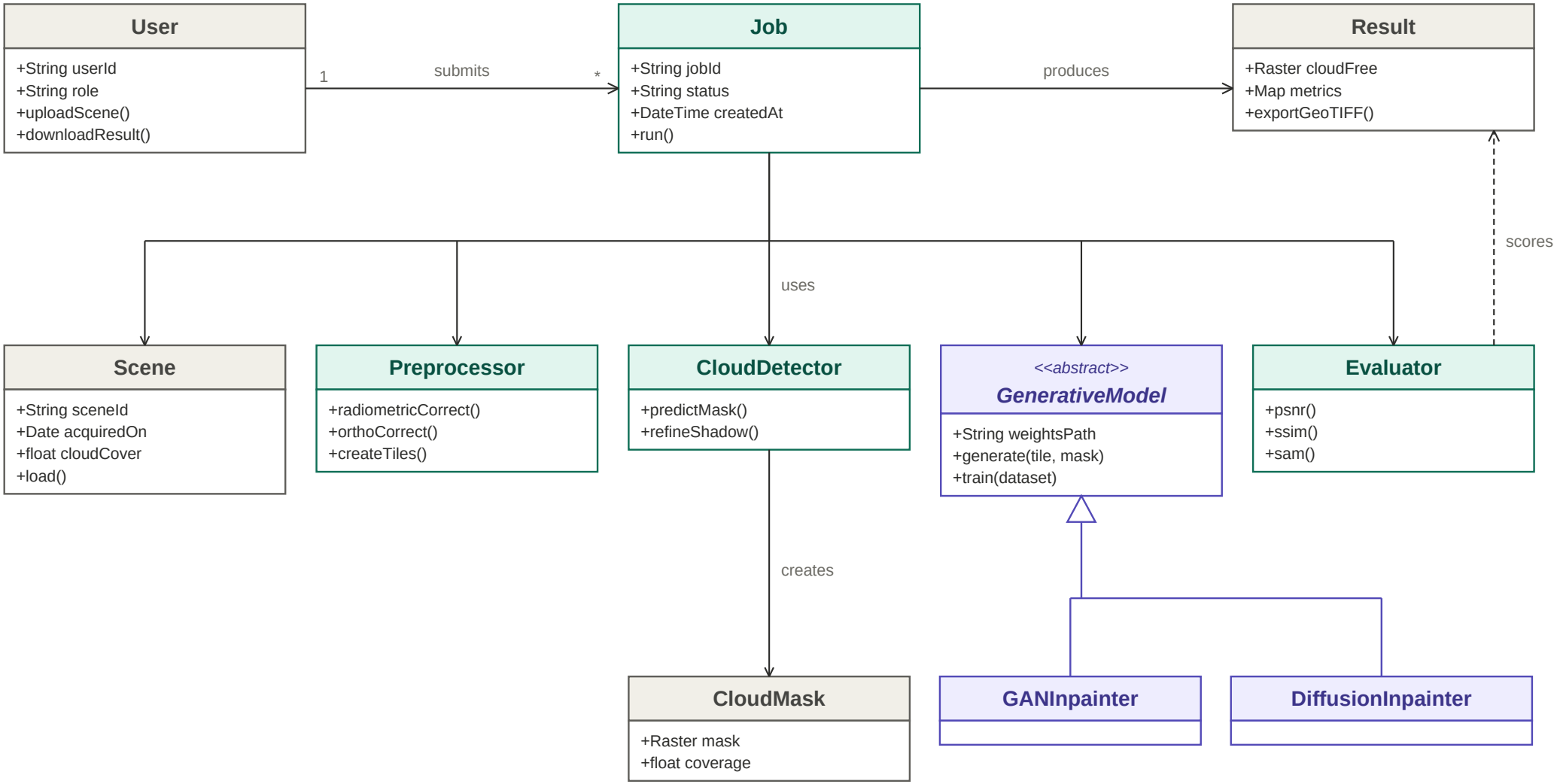


FIGURE 2

Class Diagram

Main classes with attributes and operations. GenerativeModel is abstract, so a GAN or a diffusion model can be plugged in without changing the rest of the pipeline.



Solid arrow: association Dashed arrow: dependency Hollow triangle: inheritance (generalisation)

FIGURE 3

Sequence Diagram

Order of messages when an analyst submits one LISS-IV scene. Dashed arrows are return messages.

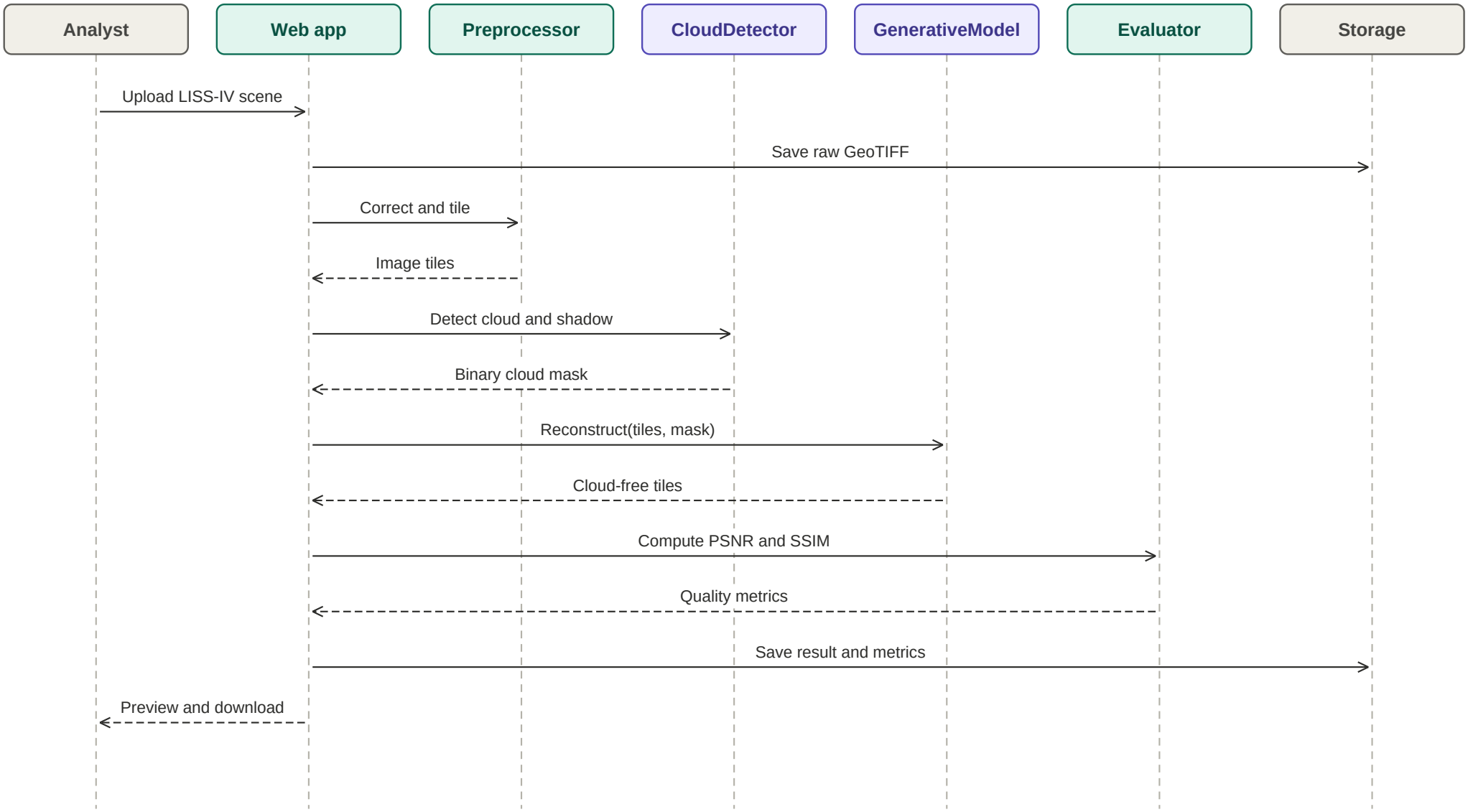


FIGURE 4

Activity Diagram

Workflow of the cloud removal pipeline, with a decision on whether cloud cover needs reconstruction and a retry loop when quality is below the target.

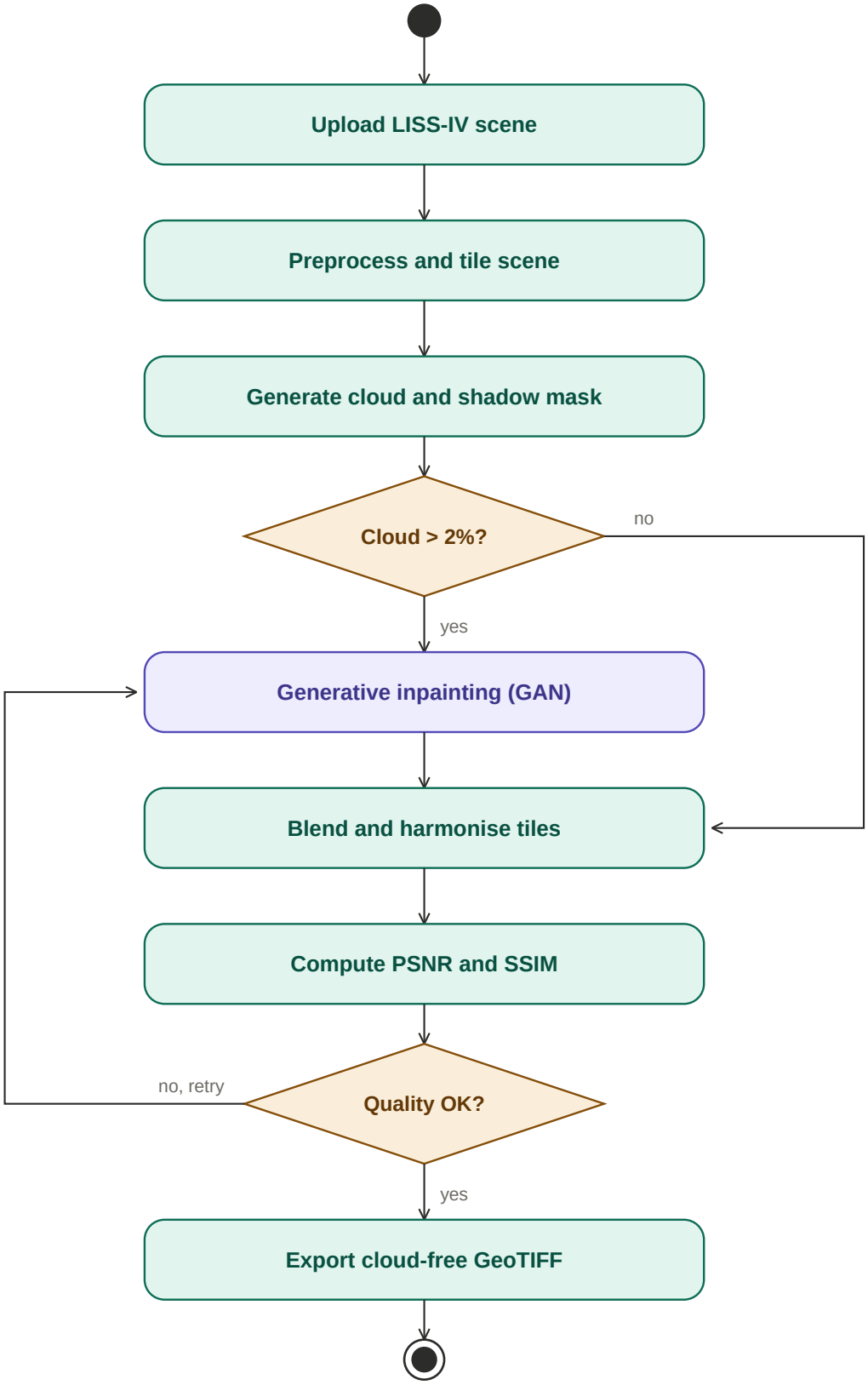
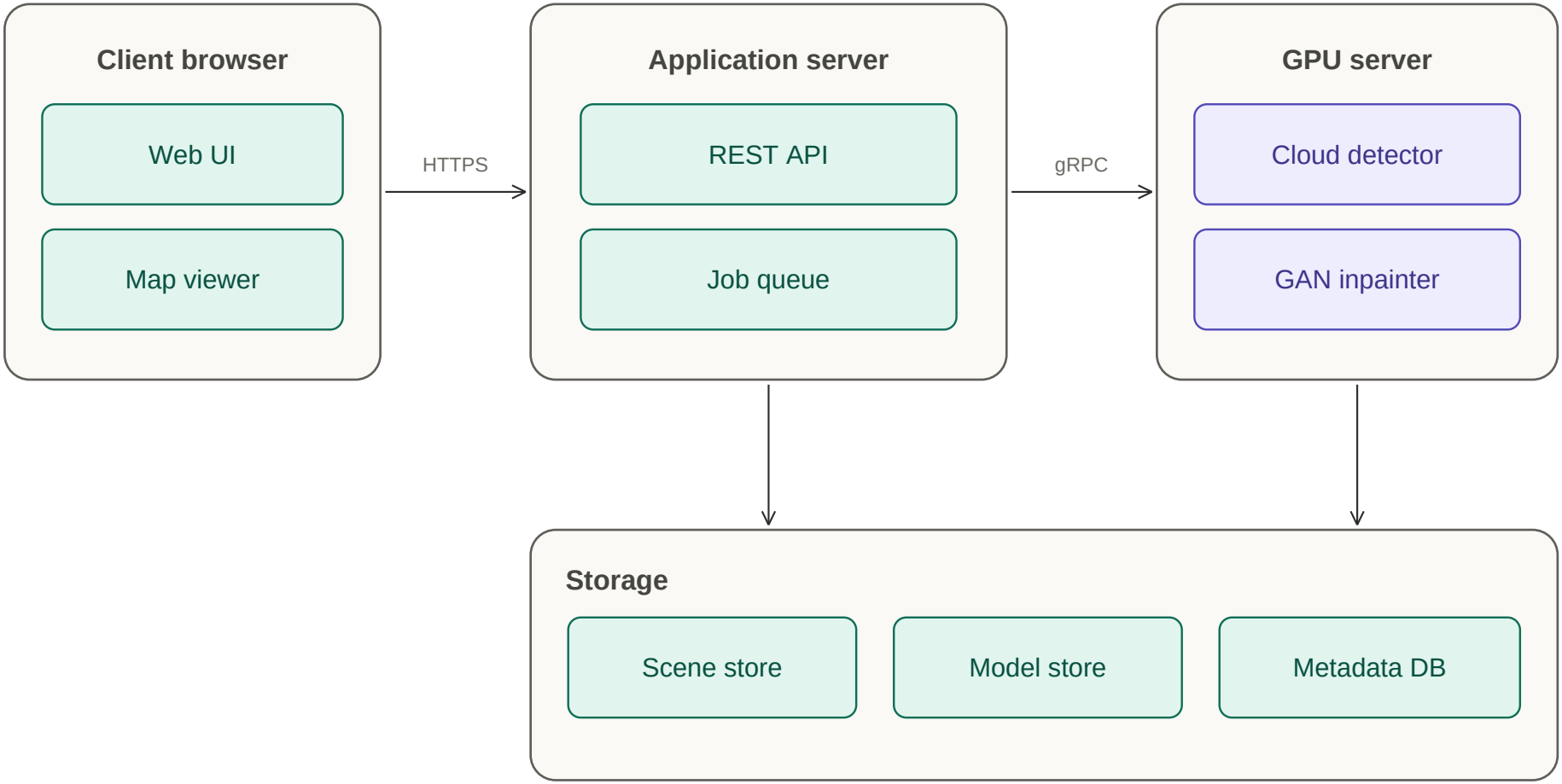


FIGURE 5

Deployment Diagram

Where each software component runs: user browser, application server, GPU server for model inference, and the storage node.